

ACID BASE BALANCE

RULE OF THE B'S

1. If the pH and the Bicarb are in the same direction → **METABOLIC**
2. If the pH and the Bicarb are in different directions → **RESPIRATORY**

Examples:

pH 7.25 (v)
HCO₃ 20 (v)
Metabolic Acidosis

pH 7.21 (v)
HCO₃ 38 (^)
Respiratory Acidosis

pH 7.50 (^)
HCO₃ 30 (^)
Metabolic Alkalosis

pH 7.50 (^)
HCO₃ 25 (normal)
Respiratory Alkalosis (NORMAL)

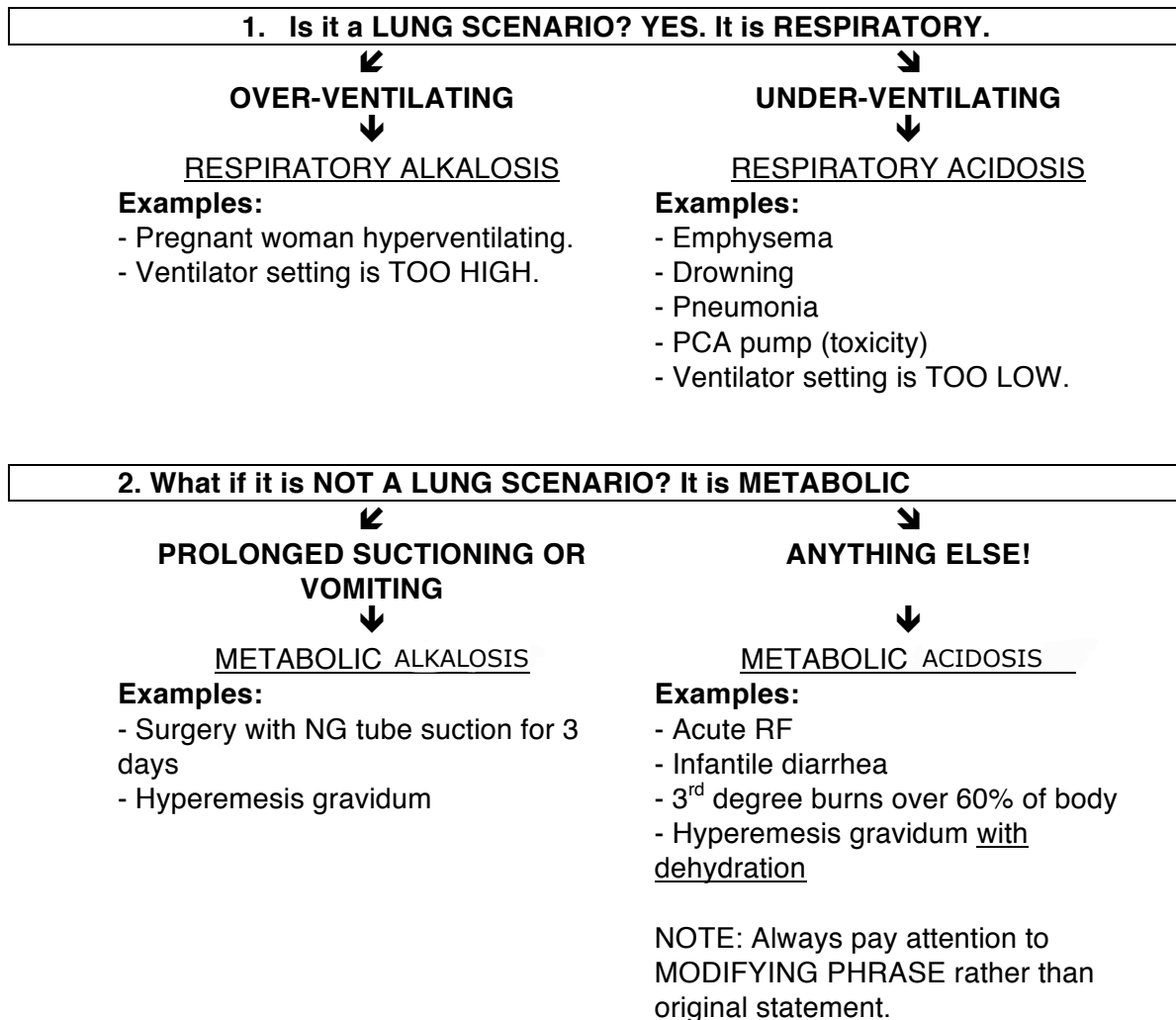
SIGNS & SYMPTOMS

Know the principles, not the lists.

PRINCIPLE: *"As the pH goes, so does my patient, except for potassium"*

<u>ALKALOSIS (^)</u>	<u>ACIDOSIS (v)</u>
Hyperreflexia Irritability Tachypnea Tachycardia Borborygmi Seizures *need suction Hypokalemia HYPOkalemia	Headache Hyporeflexia Bradycardia Bradypnea Paralytic/adynamic ileus Coma Respiratory arrest *need ambu bag MAC kussmau's (ONLY METABOLIC ACIDOSIS → "MAC") Heart block HYPERkalmia

CAUSES OF IMBALANCES



VENTILATORS AND ALARMS

HIGH PRESSURE ALARM

Ventilator is working too hard to get air into lungs. There is INCREASED RESISTANCE due to OBSTRUCTIONS.

1. Kinks → unkink
2. Water condensing into dependent loops → empty
3. Mucus in airway → turn, cough and deep breathe, suction PRN.

LOW PRESSURE ALARM

Ventilator finds that it is working too easy to get air into lungs. There is DECREASED RESISTANCE due to DISCONNECTIONS.

1. Main tubing is disconnected → reconnect
2. Oxygen sensor tubing, which senses FiO₂, is disconnected. This is the black coated wire → reconnect

Question

MD orders to disconnect ventilator in AM @ 0900hr. At 0600hr, ABC reveals respiratory acidosis. What do you do?

- a. Follow order
- b. Call MD and hold order
- c. Call RT
- d. Begin to decrease settings.

B is the answer because the patient is not able to breathe without the ventilator. The settings are TOO low. Patient should be in respiratory ALKALOSIS.

ALCOHOLISM

#1 PROBLEM: DENIAL

Psychological problem in abuse is **denial**, which is refusal to accept the reality of a problem.

Denial is the #1 problem in all abuse situations.

- It is #1 because how can you treat someone who can't admit that they have a problem?

You treat **denial by confronting** it by pointing out the difference between what they say and what they do.

Confrontation attacks the problem. Aggression attacks the person.

- You say you're not an alcohol, but it's 10AM and you already drank a 6-pack.
- You say you're not a spouse abuser, but she has a restraining order against you.

They deny, you confront.

Denial of loss + grief is different.

Stages of Grief: DABDA: Denial, anger, bargaining, depression, and acceptance

You want to **support this type of denial.**

- Guy lost one hand and wants to play piano. You do not tell him he can't. You ask him more about piano.

Pay attention to the question, is it loss or abuse?

- **With loss you support. With abuse you confront.**

#2 PROBLEM: DEPENDENCY VERSUS CODEPENDENCY

Dependency: Abuser gets significant other to do things for them. The abuser is dependent on others.

- Call in sick for me. Go buy me this. Drop me off here.

Codependency: Significant other derives positive self-esteem from making decisions for or doing things for the abuser.

- Aren't I such a great wife for calling in sick for you?
- Abuser: Life without responsibility
- Significant Other: Positive self esteem

Treatment:

- Set limits and enforce them. Teach significant other to say NO.
- Work on self-esteem of the codependent person to solve the issue.
 - I'm saying no and I'm a good person because I'm saying no.
- May solve the problem but may lose relationship.

#3 PROBLEM: MANIPULATION

Manipulation: Abuser gets significant other to do something that is not in the best interest of significant other. Nature of act is dangerous or harmful.

How is manipulation like dependency?

In both situations, the abuser is getting them to do something for them.

The difference is neutral versus negative. Look at what they're being asked to do.

If what the s/o is being asked to do is neutral = dependency/codependency.

- A 49y/o alcoholic gets her 50y/o husband to go to the store to buy alcohol for her, because this is legal.
- Last week, your sister in law calls you and says, "Would you pick up little Billy from basketball practice at school? So he can spend the night at your house b/c of the snow." You have a 4x4 Hummer and you live 3 blocks from the school, and you do it.

If what the s/o is being asked to do is harmful/dangerous to s/o = manipulation.

- A 49y/o alcoholic gets her 17y/o daughter to go to the store and buy alcohol for her, because this is illegal as she is under age.
- What if your sister in law calls you and she has the 4x4 Hummer and she lives 3 blocks from school and she asks you to pick up her son and you have a KIA that is breaking down and you live 20 miles away. This is dangerous. She can do it herself more safely.

Treatment:

- Set limits and enforce them. You say NO.
- Easier to treat because nobody likes being manipulated.
- No positive self-esteem issue with manipulation like there is with codependency/dependency.

How many patients do you have with DENIAL? 1.

How many patients do you have with DEPENDENCY/CODEPENDENCY? 2.

How many patients do you have with MANIPULATION? 1.

WERNICKE-KORSAKOFF SYNDROME – encephalopathy psychosis

- Psychosis induced by Vitamin B1/thiamine deficiency
- Lose touch with reality
- Psychotic people
- Primary symptom: amnesia with confabulation (memory loss, making up stories) because they forgot and **they believe it**
- Memory loss is "what happened in the 1990's?" not "I got drunk last night, what happened?" so they make up stories of what they were doing in the 90's

Treatment:

- Do not present reality because they won't learn it.
- **Redirect** → rechannel it into something they can do not telling them what they can't do:
 - Pt: "I'm going to presidential meeting at 8AM?"
 - Well why don't we take a shower then watch CNN and watch what's going on in Washington?

Characteristics:

- Preventable → take vitamin B1 which is a coenzyme needed for metabolism for alcohol; if deficient, alcohol will be stored and ruin brain cells.
- Arrestable → They don't have to stop drinking; they just have to take vitamin B1.
 - Stop it from getting worse? Take vitamin B1
- Irreversible → 70% irreversible (on boards, answer with majority).

ANTABUSE ARABIA (DISULFIRAM)

- Aversion therapy
- Aversion: Really strong hatred for something; gut hatred for something; we want alcoholics to really hate alcohol
- Works in theory better than in real life
- Onset and Duration: Two weeks before effectiveness and two weeks off drug until they can drink again

Patient Teaching

- Avoid all forms of alcohol to avoid nausea, vomiting, death including:
- Mouthwash even if they swish and spit
- Aftershave even if they put it on topically b/c causes nausea
- Perfumes and colognes for the same reason
- Insect repellants
- Any OTC that ends in -elixer
- Alcohol based hand sanitizers
- Uncooked icings because they have vanilla extract
- They CAN have red wine vinaigrette

OVERDOSE AND WITHDRAWAL

QUESTION 1: Every abused drug is either an upper or a downer. Therefore, figure out if the drug is an UPPER or LOWER drug.

What is the #1 most abused class of drug that is not an upper or downer? Laxatives in the elderly.

When you get a question, establish if it's an upper or a downer?

UPPERS: THERE ARE FIVE

- Caffeine
- Cocaine
- PCP/LSD (psychedelic hallucinogen)
- Methamphetamines (crystal meth, pseudopheds, etc.)
- Adderol (ADD drug)

SIGNS & SYMPTOMS: Make things go up; increased

- Euphoria
- Tachycardia
- Tachypnea
- Restlessness
- Hypertension
- Irritability
- Diarrhea
- Borborygmi
- 3+/4+ reflexes
- Spastic
- Seizures → suction @ bedside

DOWNERS: THERE ARE 135. Everything that's not an UPPER. So ONLY memorize the UPPERS!!!

SIGNS & SYMPTOMS: Make things go down; decreased

- Respiratory arrest is the biggest risk
- All symptoms above but opposite

Example: Patient is high on cocaine. What is critically important to assess?

- A. Reflexes
- B. Make sure respiratory rate is above 12
- C. HR
- D. Bowel sounds

Not B, because it's not a downer. It's an UPPER! Do not just use ABC's.

QUESTION 2: After you distinguish whether the drug is upper or downer, the second thing you ask is, are they talking about overdose or withdrawal?

Overdose/intoxication: You have too much.

Overdosed on upper → everything goes UP → pick the UP things.
Intoxication on downer → everything goes DOWN → pick the DOWN things.

Withdrawal: You don't have enough

Withdrawal downer → you don't have enough downer → pick the UP things.

Withdrawal upper → you don't have enough upper → pick the DOWN things.

Upper overdose → Downer withdrawal
Downer overdose → Upper withdrawal

Which two situations would respiratory arrest be of concern?

Downer overdose

Upper withdrawal

Which two situations would seizure be biggest risk?

Upper overdose

Downer withdrawal

Example: Bringing patient who is overdosed on cocaine. What would you expect to see, SATA.

1. Irritability

2. 4+ reflexes

3. Resp less than 12

4. Difficult to arouse

5. Borborygmi

6. Increased temp

Thought Process:

- Patient is on UPPER drug.
- Overdose on UPPER.
- Therefore TOO MUCH UPPER.
- This is a CNS drug not an AUTONOMIC.

DRUG ABUSE IN THE NEWBORN

Always **assume intoxication** not withdrawal at birth.

Within 24 hours = intoxication

After 24 hours = withdrawal

Example: Caring for infant born to a quaalude addicted mom. 24 hours after birth, SATA.

- 1. Difficult to console**
2. Low core body temp
- 3. Exaggerated startle reflex**
4. Respiratory depression
- 5. Seizure risk**
- 6. Shrill high-pitched cry**

Thought Process:

- Downer drug
- Withdrawal of downer/Overdose on upper, since it's 24 hours
- Too much UPPER

ALCOHOL WITHDRAWAL SYNDROME VERSUS DELIRIUM TREMONS

Not the same thing!

A. Every alcoholic goes through alcohol withdrawal 24 hours after they stop drinking. Only a minority (under 20%) get delirium tremens.

- Occurs 72 hours after.
- Alcohol withdrawal always comes first within 24 hours.

Alcohol withdrawal syndrome always precedes delirium tremens; however, delirium tremens does not always follow alcohol withdrawal syndrome.

B. AWS is not life threatening. DT can kill you.

C. Patients with AWS are not a danger to self or others. Patients with DT are dangerous to self and others.

AWS	DT
Regular diet.	NPO or clear liquids. Because of seizure because withdrawing from downer → UP S&S
Semi-private anywhere on unit	Private room near nurses station b/c dangerous and unstable. Usually on step down unit.
Up ad lib. Go around anywhere they want to go.	Strict bed rest. No washroom privileges.
No restraints (not a danger).	Restraints needed. Certain restraints are futile: Soft restraints not safe/strong enough. Four point restraints are not safe enough. Restraints appropriate: Vest or two point locked leathers (opposite arm and opposite leg). Rotate q2h. What would you do first? Lock the LEG first then the ARM, then release.
BOTH RECEIVE: Antihypertensive: everything going up (withdrawal of downer) Tranquillizer Multivitamin containing Vitamin B1 to prevent WERNICKE KORSAKOFF	